Indiana Water Quality Dataset Cleaning Log

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Column	Issue	Action Taken	Justification	Impact
All Columns	Blank cells and missing values present throughout dataset	Converted blanks to NA during import	Ensures R correctly recognizes missing data and allows consistent analysis	Prevents errors in calculations and summaries
ActivityEndDate, ActivityEndTime/Time, ActivityEndTime/TimeZoneCode	197,999 missing values	Retained as NA	End date/time was not recorded for many sampling events and appears nonessential to the project objectives	Preserves dataset structure without introducing artificial values
ActivityRelativeDepthName and related depth fields	Approximately 202,000 to 203,895 missing values	Retained as NA	Most surface-water samples did not contain depth-specific informatio	Prevents introduction of misleading values

			n; missingness is structural rather than erroneous	
ProjectIdentifier and ProjectName	79,080 and 203,895 missing values respectively	Retained as NA	Project metadata are administrative fields not required for water-quality analysis	No effect on nutrient or water-quality calculations
ActivityCommentText	53,624 missing values	Retained as NA	Absence of comments does not indicate an error	Preserves original reporting conditions
ResultMeasureValue	86,167 missing values	Retained as NA and not replaced with zero	Missing values often correspond to non-detects, categorical observations, or unreported measurements	Prevents distortion of concentration statistics
ResultDetectionConditionText	117,728 missing values	Created Detection Flag variable to	Non-detect information is analyticall	Maintains environmental meaning of

		distinguish detected and non-detected observations	y important and should be preserved separately	censored measurements
ResultCommentText	166,341 missing values	Retained as NA	Comments are optional and only present when additional information is needed	Missing comments do not affect quantitative analyses
LaboratoryName	32,998 missing values	Retained as NA	Laboratory information is metadata and not required for most analyses	Maintains provenance information where available
AnalysisStartDate	56,527 missing values	Retained as NA	Samples lacking analysis dates still contain valid measurement data	Allows retention of otherwise usable observations
DetectionQuantitationLimit variables	73,927 missing values	Retained as NA	Detection limits are only reported where relevant	Preserves detection information without creating

				artificial values
Character Columns	Inconsistent capitalization and spacing	Trimmed whitespace and standardized text formatting	Improves consistency in grouping and filtering operations	Reduces duplicate categories caused by formatting differences
ActivityStartDate	Date formatting inconsistency	Converted to YYYY-MM-DD format	Creates a standard temporal format for analysis	Enables sorting and time-series analysis
All Records	Potential duplicate rows	Removed exact duplicate records using `distinct()`	Duplicate observations could bias descriptive statistics and summaries	Improves data accuracy and reliability
ResultMeasureValue	11,413 observations identified as statistical outliers using IQR	Reviewed and retained	Water-quality datasets commonly contain legitimate extreme values associated with environmental events; automatic removal	Preserves potentially important environmental signals

			could discard valid observations	
ActivityDepthHeightMeasure/MeasureValue	378 IQR outliers identified	Retained	Depth measurements may vary substantially based on site conditions and sampling methods	Prevents removal of valid field measurements
DataQuality/PrecisionValue	18 IQR outliers identified	Retained and flagged for review	Very small number of unusual values that may reflect laboratory reporting differences	Minimal effect on overall analysis
DetectionQuantitationLimitMeasure/MeasureValue	22,810 IQR outliers identified	Retained	Detection limits vary widely among analytes and analytical methods, making statistical outliers expected	Preserves laboratory reporting information

ResultCommentText	Some comments appeared concatenated	Retained original text	Qualitative metadata were not used in quantitative analyses and alteration risked changing original meaning	No impact on numerical results
ResultMeasure/MeasureUnitCode	Multiple reporting units present	Units reviewed and retained	Units represent different analytes and measurement types; universal conversion was not scientifically appropriate	Maintains measurement integrity
Numeric Variables	Checked for impossible values	No impossible values identified during inspection	All reported values appeared consistent with expected water-quality measurements and	Increased confidence in dataset reliability

			laboratory reporting practices	